

Group8Negotiator: A Behavior-Adaptive Persuasion Agent for Human-Agent Negotiation

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Abstract

We describe Group8Negotiator, our submission to the ANAC HAN 2026 league. The agent adopts a hybrid architecture in which a deterministic Boulware time-based bidding strategy and a frequency-based opponent model handle all strategic decisions, while a local large language model (Qwen3:4b-instruct, served through Ollama) generates the natural-language messages exchanged with the counterpart. The novelty of the agent is *Behavior-Adaptive Persuasion* (BAP): each opponent move is classified using the taxonomy of Hindriks et al. [2], the inferred behavior profile is fed into a time-and-behavior-adaptive acceptance threshold, and the same profile is injected into the prompt context given to the language model so that the tone of generated text adapts to the opponent’s negotiation style. Across the three official HAN scenarios (Grocery, Island, Trade) and eight reference opponents, the agent reaches agreement in 95.8% of trials, dominates fast-conceding opponents (utility up to 0.95 on the $[0, 1]$ scale), and converges to near-Nash outcomes against reactive opponents (Nash distance as low as 0.126).

Keywords: automated negotiation, multi-agent systems, opponent modeling, large language models, ANAC, HAN 2026

1 Introduction

In the HAN 2026 league, an agent negotiates over discrete multi-issue domains using the Stacked Alternating Offers Protocol augmented with natural-language text exchange. Two complementary objectives must be served simultaneously: maximizing the utility achieved at agreement against a Boulware-like resistance [1], and producing persuasive natural-language messages that are sensitive to the counterpart’s negotiation style.

Group8Negotiator addresses these objectives by cleanly separating concerns: a traditional negotiation core handles every accept / reject / propose decision, and a language-model layer generates messages that explain and frame those decisions. The two layers communicate through a shared *behavior profile* derived from the move classifier of Hindriks et al. [2]. The agent is implemented on the NegMAS platform and uses Ollama as the local LLM runtime.

2 Agent Architecture

Group8Negotiator is a subclass of `LLMMetaNegotiator`, the hybrid base provided by `negmas-llm` in which a traditional *base negotiator* produces the structural moves and the wrapping LLM produces only the accompanying text. The base negotiator in our design is `BoulwareTBNegotiator`. Around this core we attach four cooperating components, summarized below.

Bidding strategy. We inherit the standard Boulware time-based concession curve from the NegMAS library. Concession is monotonically non-increasing in the offered utility: the agent opens with its highest-utility outcome and yields slowly until close to the deadline. This standard choice gives stable behavior against a wide range of counterparts.

Opponent preference model. We implement a self-contained `FrequencyOpponentModel` in the spirit of the Smith frequency model. Issue weights are initialized uniformly; value weights are updated incrementally from the frequency of values observed in opponent offers, normalized per issue. The model is invoked both for our own move-classification computation and is registered with the NegMAS scoring layer as the agent’s `opponent.ufun` estimate, so the framework can score the accuracy of our opponent modeling.

Behavior profile (BAP novelty). At each opponent turn we compute the change in our own utility and in the estimated opponent utility between the previous and the current offer, and classify the resulting move using the taxonomy of Hindriks et al. [2] into one of six categories: CONCESSION, NICE, FORTUNATE, SELFISH, UNFORTUNATE, or SILENT. A behavior profile (COOPERATIVE / COMPETITIVE / NEUTRAL) is then inferred from the recency-weighted ratio of cooperative versus competitive moves, with exponential weights $w_i = \exp(0.5(i - K + 1))$ so that recent moves dominate the inference.

Acceptance condition. The agent accepts the current offer when its utility exceeds the adaptive threshold

$$\tau(t) = rv + (\phi(t) + \alpha(\text{profile})) \cdot (u_{\max} - rv), \quad (1)$$

where rv is the agent’s reserved value, $u_{\max} = u(\text{best outcome})$ is the maximum achievable utility, $\phi(t)$ is a phase-based fraction of the utility range that decreases monotonically with time (from 0.90 early to 0.50 in the final phase, with linear interpolation down to 0.05 in an emergency phase $t \geq 0.95$ near the deadline), and α adjusts the threshold by the inferred behavior profile (+0.05 when the opponent is cooperative, -0.05 when competitive, 0 otherwise). Expressing the threshold as a fraction of $(u_{\max} - rv)$, rather than as an absolute number, is what allows the same agent to behave consistently across utility functions normalized to $[0, 1]$ and ones whose codomain is $[0, 100]$, without any per-scenario tuning.

Communication strategy. For each move (propose, accept, or reject) the agent constructs an LLM user-message that includes the standard NegMAS context tags plus a compact *BAP context block* describing the inferred opponent profile, the three most recent move classifications, and a directive instructing the model to adapt its tone accordingly. The LLM is then queried for a short structured JSON response containing the message text. The text accompanies the structural decision (offer or response) but never overrides it.

3 Results

We evaluate the agent on the three official HAN 2026 scenarios (Grocery, Island, Trade) against eight reference opponents: the agent against itself, the three NegMAS time-based negotiators (BoulwareTB, ConcederTB, NaiveTitForTat), and four agents drawn from the HAN reference codebase (SimpleNeg, BOANeg, TemplateBasedAdapter, BoulwareBasedLLM). All negotiations use a 100-step deadline. Outcome metrics are deterministic because every decision component is rule-based; the LLM influences text but not scored decisions, so repeated runs of any pairing yield numerically identical outcome values and a standard deviation of zero across runs. Table 1 reports the per-pairing outcomes.

The agent obtains the highest individual utility against time-conceding opponents ($UA = 0.92, 0.95$ and 77 against ConcederTB across the three scenarios), thanks to BAP’s firm acceptance thresholds in the early and middle phases of the negotiation. Against the reactive NaiveTitForTat in Grocery the agent simultaneously achieves the highest social welfare (1.520) and the lowest Nash distance (0.126) of any pairing, indicating that the behavior-adaptive layer can also facilitate Pareto-efficient outcomes when the counterpart responds to concessions. Boulware-class opponents (BoulwareTB, TemplateBasedAdapter, BoulwareBasedLLM, and BOANeg in Grocery and Trade) all converge to the same symmetric equilibrium, which is the expected Pareto optimum between two tough negotiators. The single non-agreement is BOANeg in Island, where two structurally Boulware-like strategies cannot reach each other’s acceptance windows within the 100-step deadline in a near zero-sum domain; this is a property of mirror-strategy deadlock rather than of our agent in particular. Aggregated, the agent agrees in 23 of 24 pairings (95.8%).

The decision logic of the agent is composed exclusively

of arithmetic over the outcome space (utility evaluation, frequency updates, threshold computation) together with a small move-history buffer; there is no search, learning loop, or random sampling on the decision path. The agent therefore operates well within the per-move time budget specified by the competition, and remains responsive whether or not the LLM text layer is enabled.

4 Conclusion

Group8Negotiator brings together standard, well-understood components — Boulware time-based bidding, frequency-based opponent modeling, and a time-and-behavior-adaptive acceptance threshold — with a small but principled novelty: an opponent-behavior profile, derived from the Hindriks move taxonomy, is shared between the acceptance condition and the LLM prompt, so that quantitative resistance and qualitative tone are adapted in concert. The resulting agent produces consistently strong outcomes across the three official HAN scenarios, a high agreement rate, and natural-language messages that are sensitive to the counterpart’s negotiation style, while keeping every scored decision under deterministic, fully auditable control.

References

- [1] Faratin, P.; Sierra, C.; and Jennings, N. R. 1998. Negotiation Decision Functions for Autonomous Agents. *Robotics and Autonomous Systems*, 24(3-4): 159–182.
- [2] Hindriks, K. V.; Jonker, C. M.; and Tykhonov, D. 2011. Let’s Dans! An Analytic Framework of Negotiation Dynamics and Strategies. *Web Intelligence and Agent Systems: An International Journal*, 9(4): 319–335.

Table 1: Per-pairing outcomes. Utility on $[0, 1]$ scale for Grocery and Trade; on $[0, 100]$ scale for Island. U_A / U_O are the agent’s and the opponent’s utility at agreement; Product = $U_A \cdot U_O$; Welfare = $U_A + U_O$; Nash dist. is the Euclidean distance from the agreement to the Nash bargaining point.

Scenario	Opponent	U_A	U_O	Product	Welfare	Nash Dist.	Step
Grocery	Group8 (self)	0.730	0.500	0.365	1.230	0.308	85
Grocery	BOANeg	0.530	0.550	0.291	1.080	0.368	85
Grocery	SimpleNeg	0.370	0.850	0.315	1.220	0.433	91
Grocery	TemplateBasedAdapter	0.530	0.550	0.291	1.080	0.368	85
Grocery	BoulwareTB	0.530	0.550	0.291	1.080	0.368	85
Grocery	ConcederTB	0.920	0.240	0.221	1.160	0.573	34
Grocery	NaiveTitForTat	0.840	0.680	0.571	1.520	0.126	47
Grocery	BoulwareBasedLLM	0.530	0.550	0.291	1.080	0.368	85
Island	Group8 (self)	66.0	50.0	3300	116.0	9.85	85
Island	BOANeg	—	—	—	—	—	no agreement
Island	SimpleNeg	95.0	13.0	1235	108.0	56.61	49
Island	TemplateBasedAdapter	66.0	50.0	3300	116.0	9.85	85
Island	BoulwareTB	66.0	50.0	3300	116.0	9.85	85
Island	ConcederTB	77.0	29.0	2233	106.0	33.54	27
Island	NaiveTitForTat	68.0	52.0	3536	120.0	9.22	77
Island	BoulwareBasedLLM	66.0	50.0	3300	116.0	9.85	85
Trade	Group8 (self)	0.600	0.426	0.255	1.026	0.595	88
Trade	BOANeg	0.550	0.551	0.303	1.101	0.462	84
Trade	SimpleNeg	0.450	1.002	0.451	1.452	0.000	97
Trade	TemplateBasedAdapter	0.550	0.551	0.303	1.101	0.462	84
Trade	BoulwareTB	0.550	0.551	0.303	1.101	0.462	84
Trade	ConcederTB	0.950	0.250	0.237	1.200	0.903	33
Trade	NaiveTitForTat	0.550	0.551	0.303	1.101	0.462	84
Trade	BoulwareBasedLLM	0.550	0.551	0.303	1.101	0.462	84